Week 3 Basis Pursuit and Nullspace Condition

- Recall: Lasso

$$\hat{\theta}^L = \underset{\theta \in \mathbb{R}^d}{\arg\min} \left\{ \|Y - X\theta\|_2^2 + 2t\|\theta\|_1 \right\}$$

Consider the limiting case where $t \to 0$

$$\Rightarrow \hat{\theta}^{BP} = \underset{\theta : Y = X\theta}{\operatorname{argmin}} \|\theta\|_1$$

(Basis Pursuit algorithm)

Defn: Fix $X \in \mathbb{R}^{n \times d}$, $k \in \{1, 2, \dots\}$. X satisfies exact recovery property of order k if for any k-sparse $\theta^* \in \mathbb{R}^d$, $y := X\theta^*$ ← no noise!, the BP algorithm returns $\hat{\theta}^{BP} = \theta^*$

Defn: Fix X, k. X satisfies the null space condition of order k if

$$\|\delta_S\|_1 < \|\delta_{S^c}\|_1, \quad \forall \delta \in \mathcal{N}(X) \setminus \{0\}, \quad S : |S| \le k$$

↳ null space of X

Thm: For given X, k, null space condition and exact recovery of order k are equivalent!

Proof: NSC $\Rightarrow$ ERP:

assume θ^* is k-sparse with support S.

$$y = X\theta^*$$

Pick arbitrary $\theta' \in \mathbb{R}^d$, $\theta' \neq \theta^*$, $X\theta' = X\theta^*$,

want:

$$\|\theta'\|_1 > \|\theta^*\|_1.$$

Define $z := \theta' - \theta^* \neq 0$. Then $z \in \mathcal{N}(X)$.

NSC $\Rightarrow \|z_S\|_1 < \|z_{S^c}\|_1$. Therefore,

$$\begin{aligned}
\|\theta'\|_1 &= \|\theta'_S\|_1 + \|\theta'_{S^c}\|_1 \\
&= \|\theta'_S\|_1 + \|z_{S^c}\|_1 \\
&> \|\theta'_S\|_1 + \|z_S\|_1 \\
&\geq \|\theta'_S - z_S\|_1
\end{aligned}$$

$$= \|\theta^*_S\|_1$$

$$= \|\theta^*\|_1$$

as desired, so ERP ✓.

- ERP $\Rightarrow$ NSC: pick $\xi \in \mathcal{N}(X)$, $\xi \neq 0$,

Define: $\theta^* := \xi_S$ (k-sparse)

$\theta' := -\xi_{S^c}$

$\Rightarrow X\theta^* = X\theta' =: y$

ERP $\Rightarrow \|\theta^*\|_1 < \|\theta'\|_1$

$\Leftrightarrow \|\xi_S\|_1 < \|\xi_{S^c}\|_1$

$\Rightarrow$ NSC ✓

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Remark: extension of NSC & ERP and their equivalence available for certain nonconvex opt as well! eg. obj $= \|\theta\|_q$, $q \in (0, +\infty)$

- Robust recovery: $\frac{\|\hat{\theta}^{BP} - \theta^*\|_2}{\|\xi\|_2} \leq C$ ← ind. ξ and θ^*

$\|\xi\|_2$ ← deterministic noise/perturbation

Robust recovery $\Leftrightarrow$ Exact recovery for obj $= \|\theta\|_1$

$\Rightarrow$ ER for general obj

(see remark in lec notes for refs)

Prop: if $k \in \{1, 2, \cdots\}$, $k > 0$, then (k, k)-restricted eigenval condition $\Rightarrow$ null space condition of order k

Proof: Suppose that (k, κ)-RE holds, but NSC fails. Then $\exists\ \delta \in \mathcal{N}(X)$, $\delta \neq 0$, s.t.

$\|\delta_S\|_1 \geq \|\delta_{S^c}\|_1$, $|S| = k$.

$\Rightarrow \delta_S \neq 0$,

$\delta \in C_S := \left\{\delta : 3\|\delta_S\|_1 \geq \|\delta_{S^c}\|_1\right\}$

RE $\Rightarrow$ $\|X\delta\|_2^2 \geq \kappa \|\delta_S\|_2^2 > 0$

$\Rightarrow$ $X\delta \neq 0$

Contradiction!

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- NSC for Random Designs

  if $X \in \mathbb{R}^{n \times d}$, entries $i.i.d \sim \mathcal{N}(0, \frac{1}{n})$

  $\Rightarrow$ NSC w.h.p. if $k \lesssim \frac{n}{\log d}$.

Definition: Gaussian width of $K \subseteq \mathbb{R}^d$ is

$$w(K) := \mathbb{E} \sup_{x \in K} G^T x$$

where $G \sim N(0, I_d)$ "mean width"

"escape through the mesh" — unit sphere in $\mathbb{R}^d$ — $N(X)$

Thm (Gordon) Let $K \subseteq S^{d-1}$. Let V be uniformly distributed random $(d-n)$-dimensional subspace of $\mathbb{R}^d$. Assume Haar measure

$$w(K) < \sqrt{n}.$$

Then $V \cap K = \emptyset$ w.p. $\geq 1 - 2.5 \exp\left(-\frac{(n/\sqrt{n+1} - w(K))^2}{18}\right)$

- Proof idea: Gaussian process comparison inequalities;

  e.g. Slepian's inequality, Gordon's minimax inequality

Thm. If $d \geq 10$, $k \geq 1$, and $n \geq 400 k \log d$. Let $X \in \mathbb{R}^{n \times d}$, X_{ij} iid $\sim N(0, 1/n)$. Then X satisfies NSC of order k w.p. $\geq 1 - 2.5 d^{-k/18}$.

Proof: For any $S \subseteq \{1, 2, \cdots d\}$, define

$$K_S := \{g \in S^{d-1} : \|g_{S^c}\|_1 \leq \|g_S\|_1\}$$

$$K = \bigcup_{S : |S| = k} K_S$$

$\forall\, x \in K_S$, we have

$$G^T x = \underbrace{G_S^T x_S}_{①} + \underbrace{G_{S^c}^T x_{S^c}}_{②}$$

$$① \le \|G_S\|_2 \|x_S\|_2 \le \|G_S\|_2$$

$$② \overset{\text{Hölder ineq.}}{\le} \|G_{S^c}\|_\infty \cdot \|x_{S^c}\|_1$$

$$\le \|G\|_\infty \cdot \|x_{S^c}\|_1$$

$$\overset{\text{defn. } K_S}{\le} \|G\|_\infty \cdot \|x_S\|_1$$

$$\overset{\text{Cauchy–Schwarz}}{\le} \|G\|_\infty \cdot \sqrt{k}\, \|x_c\|_2$$

$$\le \sqrt{k}\, \|G\|_\infty$$

$$\Rightarrow \sup_{x \in K} G^T x = \sup_{S} \sup_{x \in K_S} G^T x$$

$$\le \sup_{|S| = k} \{\|G_S\|_2\} + \sqrt{k}\, \|G\|_\infty$$

$c\sqrt{\log d}$

$$\mathbb{E}\left[\sqrt{k}\, \|G\|_\infty\right] \le \sqrt{k}\left(\sqrt{2 \log 2d} + 1\right).$$

$$\mathbb{E}\left[\sup_{|S|=k} \|G_S\|_2\right]$$

Fubini's theorem

$$= \int_0^\infty \mathbb{P}\left[\sup_{|S|=k} \|G_S\|_2 > \lambda\right] d\lambda$$

change of variable

$$\le \sqrt{k} + \sqrt{2k \ln d} + \int_{t=k\ln d}^\infty \mathbb{P}\left[\sup_{|S|=k} \|G_S\|_2 > \sqrt{k}+\sqrt{2t}\right] d(\sqrt{2t})$$

differential

union bound

$$\le \sqrt{k} + \sqrt{2k \ln d} + \underbrace{\binom{d}{k}}_{\le d^k} \int_{t=k\ln d}^\infty \mathbb{P}\left[\|G_S\|_2 > \sqrt{k}+\sqrt{2t}\right] d\sqrt{2t}$$

→ see lec notes

$$\le \sqrt{k} + \sqrt{2k\ln d} + d^k \int_{t=k\ln d}^\infty 2e^{-t}\, d(\sqrt{2t})$$

change of variable

$$= \sqrt{k} + \sqrt{2k\ln d} + 2d^k \int_{s=\sqrt{2k\ln d}}^\infty e^{-s^2/2}\, ds$$

$$\le \sqrt{k} + \sqrt{2k\ln d} + 2$$

Therefore, $w(K) \le 10\sqrt{k \ln d}$.

Apply Gordon's theorem with $V = N(X)$

$\Rightarrow N(X) \cap K = \emptyset$ with probability $\ge 1 - 2.5\, d^{-k/18}$

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Remark: RE is also true w.h.p. if $n \gtrsim k \log d$ for iid. Gaussian X.

Rayleigh quotient

$$\frac{\theta^T X^T X \theta}{\theta^T \theta} = \frac{\|X\theta\|_2^2}{\|\theta\|_2^2}$$

① bound the expectation using Gaussian comparison

② bound the deviation from expectation using concentration inequalities.